VI-SCAN Dental Assistant — System Architecture

1. System Overview

VI-SCAN is a voice-first dental assistant proof of concept that retrieves patient reports, converts spoken clinical observations into editable structured drafts, and supports context-aware follow-up questions. It uses synthetic patient/report data only; it is not a diagnostic system.

2. Architecture Flow

USER INPUT Streamlit UI Voice or typed request	**OPTIONAL STT** Audio transcription Editable transcript	**FASTAPI LAYER** API endpoints Input validation	**INTERPRETATION** Rules / optional LLM Intent and entities
ASSISTANT ROUTER Selects workflow	**REPORT RETRIEVAL** Patient/report matching Ambiguity handling	**CLINICAL DRAFT** Structured findings Source phrases + uncertainty	**RESPONSE** Evidence-backed result Editable output

3. Main Components

Voice / STT ui.py, stt.py	The interface accepts voice or typed input. When configured, speech-to-text converts audio into an editable transcript before processing.
LLM / Interpretation llm.py, interpreter.py	The optional LLM proposes intent, patient/report details, or dictation fields. Its output is treated as untrusted and checked by application rules.
Tool / API Layer main.py, assistant.py, retrieval.py	FastAPI exposes endpoints. The assistant routes requests to deterministic functions for patient matching, report retrieval, and follow-up handling.
Report Generation clinical_draft.py	Dictation becomes an editable structured draft with stated findings, anatomical sites, source phrases, and uncertainty indicators. No unstated findings should be added.
Conversation / Validation session.py, conversation.py, validators.py	Temporary session context supports follow-ups such as “Summarize her report.” Validation helps identify ambiguous matches and unsupported or missing information.

4. Main API Endpoints

Endpoint	Purpose
POST /transcribe	Transcribes audio into text.
POST /assistant/message	Processes the main voice/text assistant request.
POST /reports/search	Searches patient reports and handles ambiguous matches.
POST /dictation/draft	Creates a structured clinical draft from dictation.
POST /conversation/follow-up	Handles context-dependent follow-up requests.
POST /chatbot/ask	Answers questions grounded in a selected report.

5. Safety and Output

If multiple patients match, the assistant asks for clarification; if a report is missing, it reports that rather than guessing. Answers are grounded in source data where applicable. Clinical drafts remain editable and require clinician review. This synthetic-data prototype is not intended for clinical diagnosis.